THE ANSWERS

Notations:

(a) B - Bookwork

(b) E - New example

(c) A - New application

1. This is a question to check your understanding of basic principles. A very large majority of the students answered correctly.

a) i) $P(X \leq Y) = 0.2+0.1+0.1+0.1+0.2+0.1 = 0.80$ [1 - E]

$P(X < Y) = 0.1+0.1+0.2 = 0.4$ [1 - E]

ii)

x	0	1	2
$P(X=x)$	0.4	0.3	0.3

[1 - E]

y	0	1	2
$P(Y=y)$	0.3	0.3	0.4

[1 - E]

iii) $E(X) = 0\times 0.4+1\times 0.3+2\times 0.3 = 0.9$ [1 - E]

$E(Y) = 0\times 0.3+1\times 0.3+2\times 0.4 = 1.1$ [1 - E]

iv) $\text{Var}(X) = E(X^2)-E(X)^2 = 1\times 0.3+4\times 0.3-(0.9)^2 = 0.69,$ [1 - E]

$\text{Var}(Y) = 1\times 0.3+4\times 0.4-(1.1)^2 = 0.69,$ [1 - E]

$\text{Cov}(X,Y) = E(XY)-E(X)E(Y) = 1\times 0.1+2\times 0.2+2\times 0.1+4\times 0.1-0.9\times 1.1 = 0.11$ [1 - E]

$\text{Corr}(X,Y) = \frac{0.11}{\sqrt{0.69\times 0.69}} = 0.1594.$ [1 - E]

v) X and Y are correlated since $\text{Corr}(X,Y) \neq 0$. [1 - E]

Since they are correlated, they are also dependent. Dependency can also be seen from $P(X=0,Y=0) = 0.2 \neq P(X=0)P(Y=0) = 0.12$. [1 - E]

vi) The conditional probability mass function of X given that $Y = 0,1,2$ are given as follows

x	0	1	2
$P(X=x\|Y=0)$	$\frac{0.2}{0.3} = 2/3$	$\frac{0.0}{0.3} = 0$	$\frac{0.1}{0.3} = 1/3$

[1 - E]

x	0	1	2
$P(X=x\|Y=1)$	$\frac{0.1}{0.3}=1/3$	$\frac{0.1}{0.3}=1/3$	$\frac{0.1}{0.3}=1/3$

[1 - E]

x	0	1	2
$P(X=x\|Y=2)$	$\frac{0.1}{0.4}=1/4$	$\frac{0.2}{0.4}=1/2$	$\frac{0.1}{0.4}=1/4$

[1 - E]

vii) The conditional expectation of X given that $Y=0,1,2$ are given as follows

$E(X|Y=0)=0\times 2/3+1\times 0+2\times 1/3=2/3$ [1 - E]
$E(X|Y=1)=0\times 1/3+1\times 1/3+2\times 1/3=1$ [1 - E]
$E(X|Y=2)=0\times 1/4+1\times 1/2+2\times 1/4=1$ [1 - E]

viii) $E(X)=E(E(X|Y))=2/3\times 0.3+1\times 0.3+1\times 0.4=0.9$ [2 - E]

b) i) $P(Y\leq e)=P(X\leq \log e)=P(X\leq 1)=0.5$ [2 - A]

ii) With $\mu=1$ and $\sigma^2=4$, we can write

$$F_Y(y)=P(X\leq \log y)=\int_{-\infty}^{\log y}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

if $y>0$ and 0 otherwise, and differentiate with respect to y such that

$$f_Y(y)=\frac{1}{y\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{\log y-\mu}{\sigma}\right)^2}=\frac{1}{2y\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{\log y-1}{2}\right)^2}$$

if $y>0$, and 0 otherwise. [3 - A]

2. Compared to the first question, a larger portion of the students had troubles answering this question. Part a was mostly fine, though the most common mistake is to get the wrong domain for u and v in ii and iii. Part b ii of the question was not answered well despite the fact it directly comes from the lecture notes.

a) i) Since X,Y are independent, $f_{X,Y}(x,y) = f_X(x)f_Y(y)$. [2 - A]
We can then write

$$f_{X,Y}(x,y) = \lambda^2 e^{-\lambda(x+y)}$$

for $x > 0, y > 0$ and 0 otherwise. [2 - A]

ii) With the change of variables $X = V$ and $Y = V\left(\frac{1-U}{U}\right)$, we can calculate $\det(J) = -V/U^2$, [2 - A]
such that

$$f_{U,V}(u,v) = \lambda^2 e^{-\lambda\frac{v}{u}}\frac{v}{u^2}$$

for $0 < u < 1$ and $v > 0$ and 0 otherwise. [2 - A]

iii) The marginal of U is obtained as

$$f_U(u) = \int_0^{+\infty} f_{U,V}(u,v)dv$$

$$f_U(u) = \frac{\lambda}{u}\int_0^{+\infty} \lambda\frac{v}{u}e^{-\lambda\frac{v}{u}}dv = \frac{\lambda}{u}\frac{u}{\lambda} = 1$$

for $0 < u < 1$ and 0 otherwise. [2 - A]
The marginal of V is obtained as

$$f_V(v) = \int_0^1 f_{U,V}(u,v)du$$

$$f_V(v) = \lambda^2\int_0^1 \frac{v}{u^2}e^{-\lambda\frac{v}{u}}du = \lambda e^{-\lambda v}$$

for $v > 0$ and 0 otherwise. [2 - A]

iv) U and V are not independent since $f_{U,V}(u,v) \neq f_U(u)f_V(v)$. [3 - A]

b) i) We can proceed as follows

$$\begin{aligned} E_X[E[Y|X]] &= \int_{-\infty}^{+\infty} E[Y|X=x]f_X(x)dx \text{ [2 - B]} \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} y f_{Y|X}(y|x)dy f_X(x)dx \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} y f_{X,Y}(x,y)dxdy \text{ [2 - B]} \\ &= E(Y) \end{aligned}$$

ii) We can proceed as follows

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(E(X|Y=y)-E(X))(x-E(X|Y=y))f_{X,Y}(x,y)\,dx\,dy$$
$$=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(E(X|Y=y)-E(X))(x-E(X|Y=y))f_{X|Y}(x|y)f_Y(y)\,dx\,dy \text{ [2 - B]}$$
$$=\int_{-\infty}^{+\infty}\underbrace{\left[\int_{-\infty}^{+\infty}(x-E(X|Y=y))\,f_{X|Y}(x|y)dx\right]}_{\int_{-\infty}^{+\infty}xf_{X|Y}(x|y)dx-E(X|Y=y)\int_{-\infty}^{+\infty}f_{X|Y}(x|y)dx}(E(X|Y=y)-E(X))\,f_Y(y)\,dy \text{ [2 - B]}$$
$$=\int_{-\infty}^{+\infty}[E(X|Y=y)-E(X|Y=y)]\,(E(X|Y=y)-E(X))\,f_Y(y)\,dy \text{ [2 - B]}$$
$$=0$$